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|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |          | <p>Current <math>[I] \rightarrow 1</math> SI base unit</p> <p>Force = <math>[M] \times [L] \times [T]^{-2} \rightarrow 3</math> SI base units</p>                                                                                                                                                                                                                                                                                                                                                             |
| 2 | <b>C</b> | <p>An error is systematic if repeating the measurement under the same conditions yields readings with error of the <b>same magnitude</b> and <b>sign</b>. i.e. <u>ALL</u> measurements are <u>either</u> bigger or smaller than the true value consistently. Readings with systematic error change in a <b>predictable</b> manner depending on the conditions.</p> <p>So to reduce systematic error, the source of error has to be identified and removed. Zeroing an instrument is one of such examples.</p> |
| 3 | <b>C</b> | <p> jumper at rest when <math>v = 0</math>; jumper with zero acceleration when gradient of</p>                                                                                                                                                                                                                                                                                                                                                                                                                |